

1	(a) (gravitational) force proportional to product of masses and inversely proportional to square of separation <i>either</i> point masses <i>or</i> particles <i>or</i> 'size' $\ll$ separation	M1 A1	[2]
	(b) gravitational force provides the centripetal force	B1	
	<i>either</i> $GMm/x^2 = mx\omega^2$ <i>or</i> mv^2/x <i>either</i> $\omega = 2\pi/T$ <i>or</i> $v = 2\pi x/T$ and working to $GM = 4\pi^2 x^3/T^2$	M1 A1	[3]
	(c) <i>either</i> use of gradient of graph <i>or</i> line through origin so can use single point <i>or</i> line shown extrapolated to origin	B1	
	gradient = $(4.5 \times 10^{14})/0.35$ $6.67 \times 10^{-11} \times M = 4\pi^2 \times (4.5 \times 10^{14} \times 10^9)/(0.35 \times \{24 \times 3600\}^2)$		
	correct conversion for km^3 and power of 10	C1	
	correct conversion for day^2	C1	
	$M = 1.02 \times 10^{26} \text{ kg}$	A1	[4]